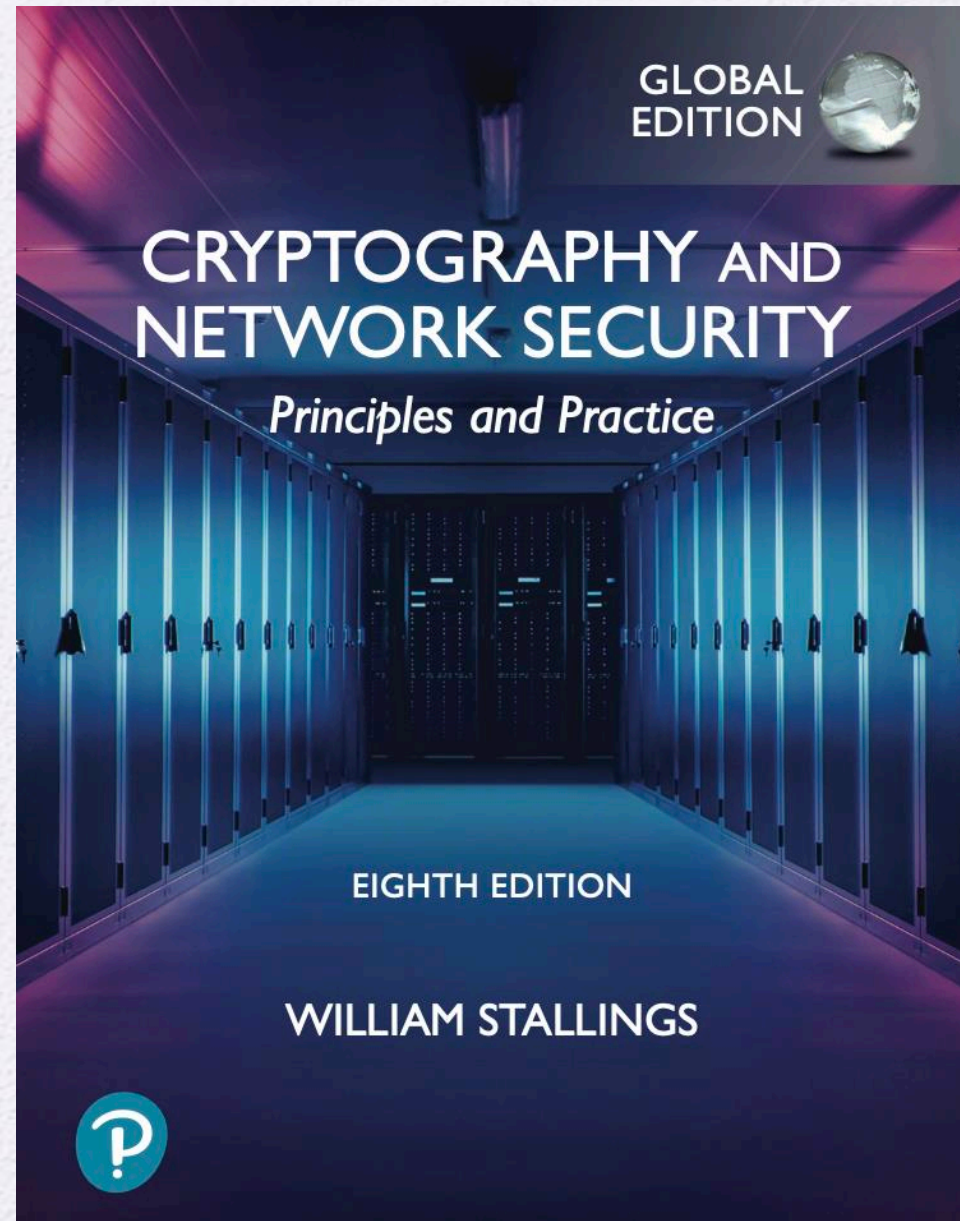


University of Nevada – Reno
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CS454/654 Reliability and Security
of Computing Systems - Fall 2024



INTRODUCTION TO NUMBER THEORY

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Divisibility

Nonzero b **divides** a if $a = mb$ for some m , where a , b and m are integers. b divides a if there is no remainder on division.

$$a \% b = 0$$

Notation: $b \mid a$ or b is a **divisor** of a .

- Positive divisors of 24 are 1, 2, 3, 4, 6, 8, 12, 24

- - If $a \mid 1$, then $a = \pm 1$.
 - If $a \mid b$ and $b \mid a$, then $a = \pm b$.
 - Any $b \neq 0$ divides 0. ^{o/o} DNE
 - If $a \mid b$ and $b \mid c$, then $a \mid c$:

→ $a \mid b$, $b \mid c$, thus $a \mid c$

$$11 \mid 66 \text{ and } 66 \mid 198 \rightarrow 11 \mid 198$$

Divisibility

Properties Continued :

- If $b|g$ and $b|h$, then $b|(mg + nh)$ for arbitrary integers m and n .

To see this last point, note that

- If $b|g$, then g is of the form $g = b \times g_1$ for some integer g_1 .
- If $b|h$, then h is of the form $h = b \times h_1$ for some integer h_1 .

So

$$mg + nh = mbg_1 + nbh_1 = b \times (mg_1 + nh_1)$$

and therefore b divides $mg + nh$.

$$b = 7; g = 14; h = 63; m = 3; n = 2$$

$$7|14 \text{ and } 7|63.$$

$$\text{To show } 7|(3 \times 14 + 2 \times 63),$$

$$\text{we have } (3 \times 14 + 2 \times 63) = 7(3 \times 2 + 2 \times 9),$$

and it is obvious that $7|(7(3 \times 2 + 2 \times 9))$.

Demo:

$$b = 4, g = 16, h = 24$$

$$m = 2, n = 3$$

$$4|16 \quad \& \quad 4|24 \quad \checkmark$$

$$4|(2(16) + 3(24))$$

$$4|4(2(4) + 3(6))$$

4 is a factor of $mg + nh$,
so it is then divisible by 4.

Division Algorithm

Given any positive integer n and any nonnegative integer a , if we divide a by n , we get any integer quotient q and an integer remainder r that follow the following relationship:

$$a = qn + r \quad 0 \leq r < n; q = \lfloor a/n \rfloor \quad (2.1)$$

where $\lfloor x \rfloor$ is the largest integer less than or equal to x . Equation (2.1) is referred to as the division algorithm.¹

$$n=15, a=2 \rightarrow q=\lfloor 2/15 \rfloor=0, \quad 2 = \overset{a}{\cancel{0}} \overset{n}{\cancel{15}} + r \rightarrow r=2$$

lower int bound

$$0 \leq 2 \leq 15 \quad \checkmark$$

Division Algorithm

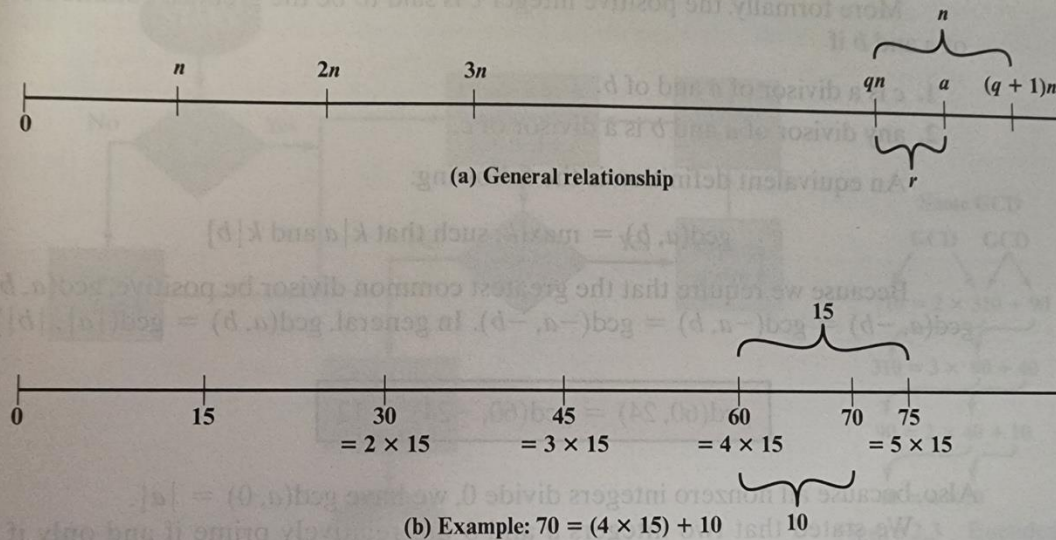


Figure 2.1 The Relationship $a = qn + r$; $0 \leq r < n$

Figure 2.1a demonstrates that, given a and positive n , it is always possible to find q and r that satisfy the preceding relationship. Represent the integers on the number line; a will fall somewhere on that line (positive a is shown, a similar demonstration can be made for negative a). Starting at 0, proceed to $n, 2n$, up to qn , such that $qn \leq a$ and $(q + 1)n > a$. The distance from qn to a is r , and we have found the unique values of q and r . The remainder r is often referred to as a **residue**.

Euclidean Algorithm

- Procedure to determine the greatest common divisor of two positive integers.
- Two integers are **relatively prime** if and only if their only common positive integer factor is 1.
- **Greatest Common Divisor (GCD):**
 - Largest integer that divides both a and b ; $\gcd$
 $(0, 0) = 0$.

Euclidean Algorithm

Positive integer c is said to be the greatest common divisor of a and b if

1. c is a divisor of a and of b .
2. Any divisor of a and b is a divisor of c .

Equivalent: $\gcd(a, b) = \max [k, \text{ such that } k \mid a \text{ and } k \mid b]$

We gcd to be positive, hence:

$$\gcd(a, b) = \gcd(a, -b) = \gcd(-a, b) = \gcd(-a, -b) \rightarrow \gcd(|a|, |b|).$$

Finding GCD

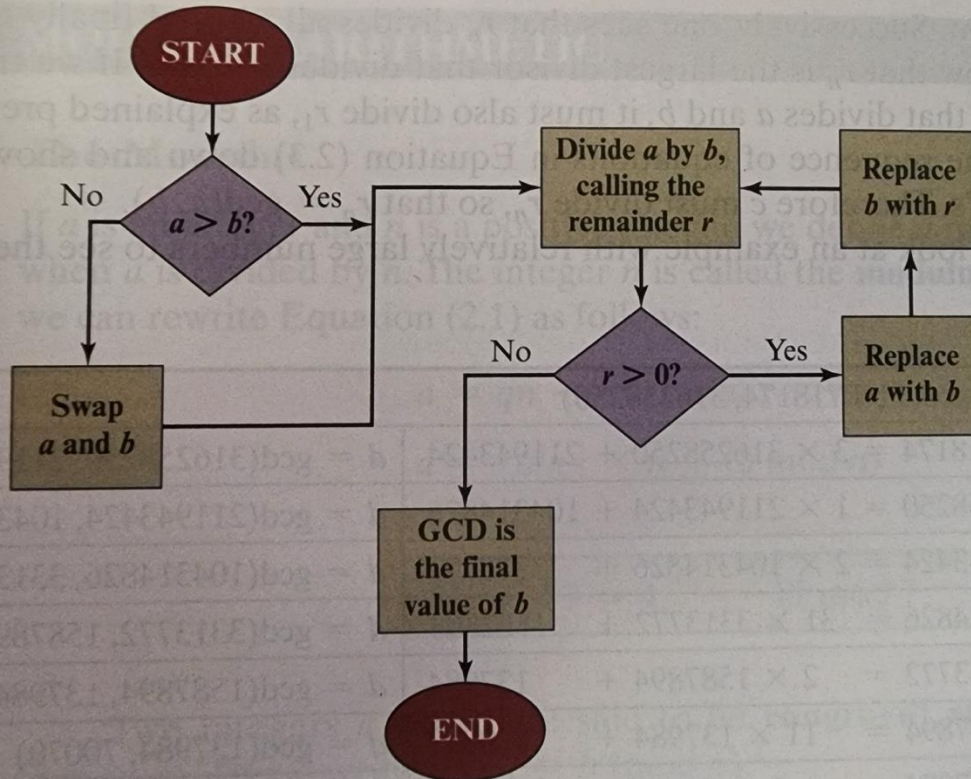
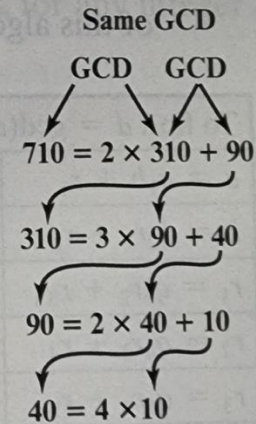


Figure 2.2 Euclidean Algorithm



$$\begin{aligned} a &= i_1 \times b + r_1 \\ b &= i_2 \times r_1 + r_2 \\ r_1 &= i_3 \times r_2 + r_3 \\ &\vdots \\ r_n &= i_{n+2} \times r_{n-1} \end{aligned}$$

Figure 2.3 Euclidean Algorithm Example: $\text{gcd}(710, 310)$

Finding GCD

Finding the Greatest Common Divisor

We now describe an algorithm credited to Euclid for easily finding the greatest common divisor of two integers (Figure 2.2). This algorithm has broad significance in cryptography. The explanation of the algorithm can be broken down into the following points:

1. Suppose we wish to determine the greatest common divisor d of the integers a and b ; that is determine $d = \gcd(a, b)$. Because $\gcd(|a|, |b|) = \gcd(a, b)$, there is no harm in assuming $a \geq b > 0$.
2. Dividing a by b and applying the division algorithm, we can state:

$$a = q_1b + r_1 \quad 0 \leq r_1 < b \quad (2.2)$$

3. First consider the case in which $r_1 = 0$. Therefore b divides a and clearly no larger number divides both b and a , because that number would be larger than b . So we have $d = \gcd(a, b) = b$.
4. The other possibility from Equation (2.2) is $r_1 \neq 0$. For this case, we can state that $d|r_1$. This is due to the basic properties of divisibility: the relations $d|a$ and $d|b$ together imply that $d|(a - q_1b)$, which is the same as $d|r_1$.
5. Before proceeding with the Euclidian algorithm, we need to answer the question: What is the $\gcd(b, r_1)$? We know that $d|b$ and $d|r_1$. Now take any arbitrary integer c that divides both b and r_1 . Therefore, $c|(q_1b + r_1) = a$. Because c divides both a and b , we must have $c \leq d$, which is the greatest common divisor of a and b . Therefore $d = \gcd(b, r_1)$.

Finding GCD

Let us now return to Equation (2.2) and assume that $r_1 \neq 0$. Because $b > r_1$, we can divide b by r_1 and apply the division algorithm to obtain:

$$b = q_2 r_1 + r_2 \quad 0 \leq r_2 < r_1$$

As before, if $r_2 = 0$, then $d = r_1$ and if $r_2 \neq 0$, then $d = \gcd(r_1, r_2)$. Note that the remainders form a descending series of nonnegative values and so must terminate when the remainder is zero. This happens, say, at the $(n + 1)$ th stage where r_{n-1} is divided by r_n . The result is the following system of equations:

$$\left. \begin{array}{ll} a = q_1 b + r_1 & 0 < r_1 < b \\ b = q_2 r_1 + r_2 & 0 < r_2 < r_1 \\ r_1 = q_3 r_2 + r_3 & 0 < r_3 < r_2 \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ r_{n-2} = q_n r_{n-1} + r_n & 0 < r_n < r_{n-1} \\ r_{n-1} = q_{n+1} r_n + 0 & \\ d = \gcd(a, b) = r_n & \end{array} \right\} \quad (2.3)$$

At each iteration, we have $d = \gcd(r_i, r_{i+1})$ until finally $d = \gcd(r_n, 0) = r_n$. Thus, we can find the greatest common divisor of two integers by repetitive application of the division algorithm. This scheme is known as the Euclidean algorithm. Figure 2.3 illustrates a simple example.

Euclidian algorithm for GCD

$$r_n = q_{n+2}r_{n+1} + r_{n+2}$$

To find $d = \gcd(a, b) = \gcd(1160718174, 316258250)$		
$a = q_1b + r_1$	$1160718174 = 3 \times 316258250 + 211943424$	$d = \gcd(316258250, 211943424)$
$b = q_2r_1 + r_2$	$316258250 = 1 \times 211943424 + 104314826$	$d = \gcd(211943424, 104314826)$
$r_1 = q_3r_2 + r_3$	$211943424 = 2 \times 104314826 + 3313772$	$d = \gcd(104314826, 3313772)$
$r_2 = q_4r_3 + r_4$	$104314826 = 31 \times 3313772 + 1587894$	$d = \gcd(3313772, 1587894)$
$r_3 = q_5r_4 + r_5$	$3313772 = 2 \times 1587894 + 137984$	$d = \gcd(1587894, 137984)$
$r_4 = q_6r_5 + r_6$	$1587894 = 11 \times 137984 + 70070$	$d = \gcd(137984, 70070)$
$r_5 = q_7r_6 + r_7$	$137984 = 1 \times 70070 + 67914$	$d = \gcd(70070, 67914)$
$r_6 = q_8r_7 + r_8$	$70070 = 1 \times 67914 + 2156$	$d = \gcd(67914, 2156)$
$r_7 = q_9r_8 + r_9$	$67914 = 31 \times 2156 + 1078$	$d = \gcd(2156, 1078)$
$r_8 = q_{10}r_9 + r_{10}$	$2156 = 2 \times 1078 + 0$	$d = \gcd(1078, 0) = 1078$
Therefore, $d = \gcd(1160718174, 316258250) = 1078$		

$d = \gcd(a, b)$

where $a \geq b > 0$

$a = q_1 * b + r_1$

$0 \leq r_1 < b$

swap if needed

division algorithm

remainder

If $r_1 = 0$ **b** divides **a** and $d = \gcd(a, b) = b$

Else $\gcd(a, b) = \gcd(b, r_1) = d$

If **d** divides **a** and **b**, it must also divide any linear combination of **a** and **b** ($r = a - bq$)

Similarly, if **d** divides **b** and **r**, it must also divide any linear combination of **b** and **r**, particularly $a = bq + r$

if $\gcd(A, B) = 1$

then A and B are relatively prime

A=15 and B=8

to each other, not literally prime

Modular Arithmetic

- The modulus
 - If a is an integer and n is a **positive** integer, we define $a \bmod n$ to be the remainder when a is divided by n ; the integer n is called the **modulus**

- Thus, for any integer a :

$$a = qn + r \quad 0 \leq r < n; \quad q = [a/n]$$

$$a \bmod n = r \quad \xrightarrow{\text{modulus}} \\ \hookrightarrow a \% n = r$$

$$11 \bmod 7 = 4$$

$$-11 \bmod 7 = 3$$

$$-11 + 7k \equiv r \quad \text{where } 0 \leq r < 7$$

↙ equivalent/congruent

$$-11 + 7 \times 1 = -4 \quad (\text{still negative})$$

$$-11 + 7 \times 2 = 3 \quad (\text{in range } [0, 7))$$

Modular Arithmetic

- Congruent modulo n

$\begin{matrix} a & 7 \bmod 5 = 2 \\ \rightarrow & b & 2 \bmod 5 = 2 \end{matrix}$

$\rightarrow 2 \not\equiv 7$ are congruent modulo 5, thus $\overset{b}{2} \equiv \overset{a}{7}$, but clearly $2 \neq 7$.
- Two integers a and b are said to be **congruent modulo n** if $(a \bmod n) = (b \bmod n)$
- This is written as $a \equiv b \pmod{n} \rightarrow 7 \equiv 2 \pmod{5}$
- Note that if $a \equiv 0 \pmod{n}$, then $n \mid a$ *eg. $6 \equiv 0 \pmod{2}$, $2 \mid 6$*

$\uparrow$ no remainder r !
 $\uparrow$ 2 is a factor of 6

$$73 \equiv 4 \pmod{23};$$

$$21 \equiv -9 \pmod{10}$$

Properties of Congruences

- Congruences have the following properties:

$$\text{eg } 73 \equiv 4 \pmod{23} \rightarrow 23 \mid (73 - 4), r=0 \checkmark$$

1. $a \equiv b \pmod{n}$ if $n \mid (a - b)$.

2. $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$. $\rightarrow a \% n = b \% n$

3. $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$

$$73 \equiv 4 \pmod{23} \quad 4 \equiv 27 \pmod{23} \quad 73 \equiv 27 \pmod{23}$$

To demonstrate the first point, if $n \mid (a - b)$, then $(a - b) = kn$ for some k

- So we can write $a = b + kn$
- Therefore, $(a \bmod n) = (\text{remainder when } b + kn \text{ is divided by } n) = (\text{remainder when } b \text{ is divided by } n) = (b \bmod n)$

Modular Arithmetic

- Modular arithmetic exhibits the following properties:

1. $[(a \text{ mod } n) + (b \text{ mod } n)] \text{ mod } n = (a + b) \text{ mod } n$

2. $[(a \text{ mod } n) - (b \text{ mod } n)] \text{ mod } n = (a - b) \text{ mod } n$

3. $[(a \text{ mod } n) * (b \text{ mod } n)] \text{ mod } n = (a * b) \text{ mod } n$

simplifications

$$11 \text{ mod } 8 = 3; 15 \text{ mod } 8 = 7$$

$$[(11 \text{ mod } 8) + (15 \text{ mod } 8)] \text{ mod } 8 = 10 \text{ mod } 8 = 2$$

$$(11 + 15) \text{ mod } 8 = 26 \text{ mod } 8 = 2$$

$$[(11 \text{ mod } 8) - (15 \text{ mod } 8)] \text{ mod } 8 = -4 \text{ mod } 8 = 4$$

$$(11 - 15) \text{ mod } 8 = -4 \text{ mod } 8 = 4$$

$$[(11 \text{ mod } 8) \times (15 \text{ mod } 8)] \text{ mod } 8 = 21 \text{ mod } 8 = 5$$

$$(11 \times 15) \text{ mod } 8 = 165 \text{ mod } 8 = 5$$

Table 2.3 Properties of Modular Arithmetic for Integers in $\mathbb{Z}_n$

Property	Expression
Commutative Laws	$(w + x) \bmod n = (x + w) \bmod n$ $(w \times x) \bmod n = (x \times w) \bmod n$
Associative Laws	$[(w + x) + y] \bmod n = [w + (x + y)] \bmod n$ $[(w \times x) \times y] \bmod n = [w \times (x \times y)] \bmod n$
Distributive Law	$[w \times (x + y)] \bmod n = [(w \times x) + (w \times y)] \bmod n$
Identities	$(0 + w) \bmod n = w \bmod n$ $(1 \times w) \bmod n = w \bmod n$
Additive Inverse $(-w)$	For each $w \in \mathbb{Z}_n$, there exists a z such that $w + z \equiv 0 \bmod n$

order

grouping

we can make $r=0$

if $(a + b) \equiv (a + c) \pmod{n}$ **then** $b \equiv c \pmod{n}$

cancel

$$\boxed{(\cancel{5} + 23) \equiv (\cancel{5} + 7) \pmod{8}; 23 \equiv 7 \pmod{8}}$$

cancel

cancel

$$((\cancel{-a}) + a + b) \equiv ((\cancel{-a}) + a + c) \pmod{n}$$

$$b \equiv c \pmod{n}$$

if $(a \times b) \equiv (a \times c) \pmod{n}$ **then** $b \equiv c \pmod{n}$ **if** a is relatively prime to n

if $(a \times b) \equiv (a \times c) \pmod{n}$ then $b \equiv c \pmod{n}$ if a is relatively prime to n

To see this, consider an example in which the condition of Equation (2.5) does not hold. The integers 6 and 8 are not relatively prime, since they have the common factor 2. We have the following:

$$6 \times 3 = 18 \equiv 2 \pmod{8}$$

$$6 \times 7 = 42 \equiv 2 \pmod{8}$$

Yet $3 \not\equiv 7 \pmod{8}$.

Residue classes of (mod n) is $[0], [1], [2], \dots, [n - 1]$

above mentioned

if $(a \times b) \equiv (a \times c) \pmod{n}$ then $b \equiv c \pmod{n}$ if a is relatively prime to n

The reason for this strange result is that for any general modulus n , a multiplier a that is applied in turn to the integers 0 through $(n - 1)$ will fail to produce a complete set of residues if a and n have any factors in common.

With $a = 6$ and $n = 8$, $\gcd(6, 8) \neq 1$

Z_8	0	1	2	3	4	5	6	7
Multiply by 6	0	6	12	18	24	30	36	42
Residues	0	6	4	2	0	6	4	2

Because we do not have a complete set of residues when multiplying by 6, more than one integer in Z_8 maps into the same residue. Specifically, $6 \times 0 \pmod{8} = 6 \times 4 \pmod{8}$; $6 \times 1 \pmod{8} = 6 \times 5 \pmod{8}$; and so on. Because this is a many-to-one mapping, there is not a unique inverse to the multiply operation.

However, if we take $a = 5$ and $n = 8$, whose only common factor is 1,

Z_8	0	1	2	3	4	5	6	7
Multiply by 5	0	5	10	15	20	25	30	35
Residues	0	5	2	7	4	1	6	3

note the "pattern"

The line of residues contains all the integers in Z_8 , in a different order.

$$\begin{aligned} Z_n &= 0 \dots n-1 \\ y_n &= Z_n \times a \\ y_n \% n &= r_n \\ r_n &= Z_n \end{aligned}$$

Prime Numbers

- Prime number: any integer $p > 1$ for which only divisors are $+1, -1, +p, -p$.
 - All the numbers other than $+1, -1$, and the prime numbers are **composite numbers**.
- Composite numbers are product of at least two prime numbers.

Any integer $a > 1$ can be factored in a unique way as

$$a = p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_t^{a_t} \quad (2.9)$$

where $p_1 < p_2 < \cdots < p_t$ are prime numbers and where each a_i is a positive integer. This is known as the fundamental theorem of arithmetic; a proof can be found in any text on number theory.

$p \in P$
prime # p within the
greater set of all prime #s, P .

$$a = \prod_{p \in P} p^{a_p} \quad \text{where each } a_p \geq 0$$

π_i , for multiplication sequences. Σ sigma is for addition.

Prime Numbers

$$a = \prod_{p \in P} p^{a_p} \quad \text{where each } a_p \geq 0$$

Multiplication of two numbers is equivalent to adding the corresponding exponents. Given $a = \prod_{p \in P} p^{a_p}$, $b = \prod_{p \in P} p^{b_p}$. Define $k = ab$. We know that the integer k can be expressed as the product of powers of primes: $k = \prod_{p \in P} p^{k_p}$. It follows that $k_p = a_p + b_p$ for all $p \in P$.

$$k = 12 \times 18 = (2^2 \times 3) \times (2 \times 3^2) = 216$$

$$k_2 = 2 + 1 = 3; k_3 = 1 + 2 = 3$$

$$216 = 2^3 \times 3^3 = 8 \times 27$$

8 & 27 are relatively prime, $\gcd(8, 27) = 1$.

Prime Numbers

What does it mean, in terms of the prime factors of a and b , to say that a divides b ? Any integer of the form p^n can be divided only by an integer that is of a lesser or equal power of the same prime number, p^j with $j \leq n$. Thus, we can say the following.

Given

$$a = \prod_{p \in P} p^{a_p}, b = \prod_{p \in P} p^{b_p}$$

If $a|b$, then $a_p \leq b_p$ for all p .

$p \in P$
prime # p within the
greater set of all prime #s, P .

prime # p

$$a = 12; b = 36; 12|36$$

$$12 = 2^2 \times 3; 36 = 2^2 \times 3^2$$

$$a_2 = 2 = b_2$$

$$a_3 = 1 \leq 2 = b_3$$

Thus, the inequality $a_p \leq b_p$ is satisfied for all prime numbers.

If $k = \gcd(a, b)$, then $k_p = \min(a_p, b_p)$ for all p .

$1^n, n \geq 0$,
will always = 1.
not counted.

$$300 = 2^2 \times 3^1 \times 5^2$$

$$18 = 2^1 \times 3^2$$

$$\gcd(18, 300) = 2^1 \times 3^1 \times \cancel{5^0} = 6$$

we use lower

$$2 \vee 1 \quad 1 \wedge 2 \quad 2 \vee 0$$

$$1 \times 2 \times 3 \times 5 = 6$$

if $(a \times b) \equiv (a \times c) \pmod{n}$ **then** $b \equiv c \pmod{n}$ **if** a is relatively prime to n

if $(a \cdot b) \equiv (a \cdot c) \pmod{n}$ then $b \equiv c \pmod{n}$

if a is relatively prime to n .

a is relatively prime to n $a=5$ $n=8$

$\mathbb{Z}_8$	0	1	2	3	4	5	6	7
Multiply by 5	0	5	10	15	20	25	30	35
Residues	0	5	2	7	4	1	6	3

$(6 \cdot 5) = 30$
 $30 \pmod{8} = 6$

pattern!

a is not relatively prime to n $a=6$ $n=8$

$\mathbb{Z}_8$	0	1	2	3	4	5	6	7
Multiply by 6	0	6	12	18	24	30	36	42
Residues	0	6	4	2	0	6	4	2

if a is relatively prime to n we know that there exist integer a^{-1} for which $(a \cdot a^{-1}) \pmod{n} = 1$

for example $a=5$ $n=8$ $a^{-1}=5$
 what? a^{-1} is $\frac{1}{5}$. $a \cdot a^{-1} = 1$

$$(a \cdot b) \equiv (a \cdot c) \pmod{n}$$

$\Downarrow$ means

$$(a \cdot b) \pmod{n} = (a \cdot c) \pmod{n}$$

$\Downarrow$ then

$$a \cdot b = k_1 \cdot n + r$$

$$a \cdot c = k_2 \cdot n + r$$

because $\implies$

$$(a \cdot b) \pmod{n} = (k_1 \cdot n + r) \pmod{n} = r$$

$$(a \cdot c) \pmod{n} = (k_2 \cdot n + r) \pmod{n} = r$$

$$(a \cdot b) \equiv (a \cdot c) \pmod{n}$$

we can do

$$a^{-1} \cdot (a \cdot b) \equiv a^{-1} \cdot (a \cdot c) \pmod{n}$$

because $(a^{-1} \cdot a \cdot b) \pmod{n} = (a^{-1} \cdot a \cdot c) \pmod{n}$

$$(k_1 \cdot n \cdot a^{-1} + r \cdot a^{-1}) \pmod{n} = (k_2 \cdot n \cdot a^{-1} + r \cdot a^{-1}) \pmod{n}$$

$$(r \cdot a^{-1}) \pmod{n} = (r \cdot a^{-1}) \pmod{n}$$

we know that $(a^{-1} \cdot a) \pmod{n} = 1$ then $a^{-1} \cdot a = k_3 \cdot n + 1$

$$a^{-1} \cdot a \cdot b = k_3 \cdot n \cdot b + b \implies (a^{-1} \cdot a \cdot b) \pmod{n} = (k_3 \cdot n \cdot b + b) \pmod{n} = b \pmod{n}$$

$$a^{-1} \cdot a \cdot c = k_3 \cdot n \cdot c + c \implies (a^{-1} \cdot a \cdot c) \pmod{n} = c \pmod{n}$$

if $(a^{-1} \cdot a \cdot b) \pmod{n} = (a^{-1} \cdot a \cdot c) \pmod{n}$

then $b \pmod{n} = c \pmod{n} \implies b \equiv c \pmod{n}$

Fermat's Theorem

If p is prime and a is a positive integer not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}$$

$$3^{5-1} \equiv 1 \pmod{5}$$

$81 \pmod{5} \rightarrow 81 \% 5 = 1$

Alternative form of Fermat's theorem is also useful: If p is prime and a is just a positive integer, then

$$a^p \equiv a \pmod{p}$$

$$3^5 \equiv 3 \pmod{5}$$

$243 \pmod{5} \rightarrow 243 \% 5 = 3$ $240/5$ has $r=0$

Fermat's theorem

$$a^{p-1} \equiv 1 \pmod{p}$$

If p is prime and a is integer not divisible by p
means $\Rightarrow$ a and p is relatively prime.

because p is divided by only $1, p$
 a is divided by 1 but not p
thus $\gcd(a, p) = 1$

$P = \{1, 2, \dots, p-1\}$
multiply each element with a

$P' = \{a, 2a, \dots, (p-1)a\}$
take $\text{mod } p$ of each element to get set X

$$X = \{a \bmod p, 2a \bmod p, \dots, (p-1)a \bmod p\}$$

X will be some permutation of set P

Example $a=8$ $p=5$ a and p relatively prime

	$Z_p =$	0	1	2	3	4
multiply by 8		0	8	16	24	32
mod 5		0	3	1	4	2

multiply numbers in $P \Rightarrow 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-1) = (p-1)!$

multiply numbers in $X \Rightarrow$ should also give $(p-1)!$

~~X is~~
X is just permutation of P

$$a \bmod p \cdot 2a \bmod p \dots (p-1) \bmod p = (p-1)!$$

take mod p for both sides.

$$(a \bmod p \cdot 2a \bmod p \dots (p-1) \bmod p) \bmod p = (p-1)! \bmod p$$

using modular arithmetic property from Slide 7.

$$((a \bmod n)(b \bmod n)) \bmod n = (a \cdot b) \bmod n$$

$$(a \cdot 2a \cdot 3a \dots (p-1)a) \bmod p = (p-1)! \bmod p$$

$\Downarrow$

$$(a^{p-1} \cdot (p-1)!) \bmod p = (p-1)! \bmod p$$

We know
 $(a \cdot b) \bmod n = (a \cdot c) \bmod n$ if a and n relatively prime
then $b \bmod n = c \bmod n$

if $(p-1)!$ is relatively prime to p then $a^{p-1} \bmod p = 1 \bmod p$

$$a^{p-1} \equiv 1 \pmod{p}$$

$(p-1)!$ is relatively prime to p

$$(p-1)! = \underline{1} \cdot 2 \cdot 3 \dots (p-1)$$

$$p = \underline{1} \cdot p$$

$$\gcd((p-1)!, p) = 1$$

Fermat's Theorem

$$a^p \equiv 1 \pmod{p}$$

if p is prime, a is ~~not~~ positive integer not divisible by p

Alternative Fermat's Theorem

$$a^p \equiv a \pmod{p}$$

if p is prime, a is just positive integer.

Case 1: a is not divisible by p

$$a^{p-1} \equiv 1 \pmod{p} \quad (\text{from Fermat's theorem})$$

$$a \cdot a^{p-1} \equiv 1 \cdot a \pmod{p}$$

$$a^p \equiv a \pmod{p}$$

Case 2: a is divisible by p

$$a = kp$$

$$a^p = (kp)^p$$

↓ substitute into $a^p \equiv a \pmod{p}$

$$(kp)^p \equiv (kp) \pmod{p}$$

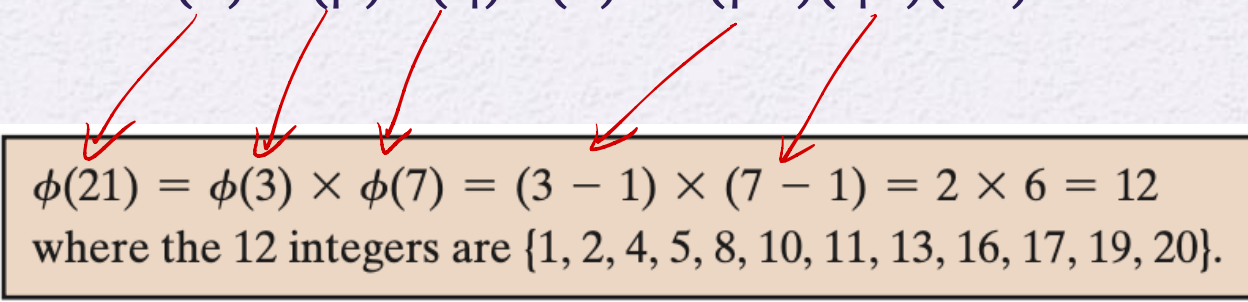
$$k^p p^p \pmod{p} = (kp) \pmod{p}$$

$$0 = 0$$

* both sides are multiples of p

Euler's Totient Function $\phi(n)$

- $\phi(n)$ is defined as the number of positive integers less than n and relatively **prime to n**
- $\phi(1) = 1$ *simple enough* *The #s in the set related to $\phi(n)$ are all + Integers*
- For prime number p : $\phi(p) = p - 1$ $\phi(17) = 16 \rightarrow 1, 2, 3, \dots, 16$
- For number n : create prime factorization
- $n = p^{e_1} q^{e_2} k^{e_3} \dots \Rightarrow \phi(n) = \phi(p) \phi(q) \phi(k) \dots = (p-1)(q-1)(k-1) \dots$



$\phi(21) = \phi(3) \times \phi(7) = (3 - 1) \times (7 - 1) = 2 \times 6 = 12$
where the 12 integers are $\{1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20\}$.

Miller-Rabin Algorithm

- Miller-Rabin algorithm is a probabilistic primality test used to determine whether a given number is prime.
- Line 1: $n-1=2^k q$ Line 5: $a^{(2^j)q} \bmod n$ Line 3: a^q
- Composite: definitely not prime
- Inconclusive: we don't know if it is prime or not

```
TEST (n)
1. Find integers  $k, q$ , with  $k > 0$ ,  $q$  odd, so that
    $(n - 1 = 2^k q)$ ;
2. Select a random integer  $a$ ,  $1 < a < n - 1$ ;
3. if  $a^q \bmod n = 1$  then return("inconclusive");
4. for  $j = 0$  to  $k - 1$  do
5.   if  $a^{2^j q} \bmod n = n - 1$  then return("inconclusive");
6. return("composite");
```


Line 1: $n-1=2^k q$

Line 5: $a^{(2^j)q} \bmod n$

Line 3: a^q

```
TEST (n)
```

```
1. Find integers  $k, q$ , with  $k > 0, q$  odd, so that  
    $(n - 1 = 2^k q)$ ;  
2. Select a random integer  $a, 1 < a < n - 1$ ;  
3. if  $a^q \bmod n = 1$  then return("inconclusive");  
4. for  $j = 0$  to  $k - 1$  do  
5. if  $a^{2^j q} \bmod n = n - 1$  then return("inconclusive");  
6. return("composite");
```

Line1: $n=29 \Rightarrow n-1=28 \Rightarrow 28=2^2 \cdot 7 \Rightarrow k=2 \quad q=7$

Line2: $1 < a < n-1 \quad a=10$

Line3: $10^7 \bmod 29 = 17$

Line5 just for $j=1$: $10^{2 \cdot 7} \bmod 29 = 28 \quad \text{return "inconclusive"}$

We can try all integers $1 < a < n-1$ and we will get "inconclusive" thus $n=29$ has high probability of being prime number.

Line 1: $n-1=2^k q$

Line 5: $a^{(2^j)q} \bmod n$

Line 3: a^q

```
TEST (n)
```

```
1. Find integers  $k, q$ , with  $k > 0, q$  odd, so that  
    $(n - 1 = 2^k q)$ ;  
2. Select a random integer  $a, 1 < a < n - 1$ ;  
3. if  $a^q \bmod n = 1$  then return("inconclusive");  
4. for  $j = 0$  to  $k - 1$  do  
5. if  $a^{2^j q} \bmod n = n - 1$  then return("inconclusive");  
6. return("composite");
```

Line1: $n = 221 = 13 * 17 \Rightarrow n-1 = 220 \Rightarrow 220 = 2^2 * 55 \Rightarrow k=2 \quad q=55$

Line2: $1 < a < n-1 \quad a=5$

Line3: $5^{55} \bmod 221 = 112$

Line4: both $j=0, j=1$ return "composite"

Thus $n=221$ is not prime, but if we try $a=21$ test returns "inconclusive".

- For odd number n that is **not prime** and randomly chosen a , $1 < a < n-1$ the probability that TEST will return inconclusive (i.e., fail to detect that n is not prime) is **less than $1/4$** .
- Thus, if we try t different values of a , then probability **becomes $(1/4)^t$**
- If $t=10$, we tried 10 different values for a , the probability of **non prime** number **passing test** is less than $(1/4)^t = 10^{-6}$
- In 2002 (announced in 2002, published in 2004), Agrawal, Kayal, and Saxena developed a relatively simple deterministic algorithm that efficiently determines whether a given large number is a prime.
- The algorithm, known as the **AKS** algorithm, **does not appear** to be as efficient as the **Miller–Rabin algorithm**.
- Thus far, AKS has not supplanted this older, probabilistic technique Miller–Rabin algorithm.